MASTER OF SCIENCE IN ENGINEERING

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Advanced Statistical Data Analysis
Worksheet Week 7

Exercise 1: We return again to the examples discussed on the Worksheets Week 4, 5, and 6.

a) **Turbine Data** (cf. Exercise 1 on Worksheet Week 4)
Does the GLM that you have fitted in part 1(b) model the data adequately?

b) **Premature Birth Data** (cf. Exercise 2 on Worksheet Week 4)
Does the logit model that you have fitted in part 2(c) model the data adequately?

c) **Dial-A-Ride Data** (cf. Exercise 2 on Worksheet Week 5)
Does the GLM that you have fitted in part 2(c) model the data adequately?

d) **Transaction Data** (cf. Exercise 4 on Worksheet Week 5)
Does the GLM that you have fitted in part 4(c) model the data adequately?

e) **Nambeware Polishing Times** (cf. Exercise 5 on Worksheet Week 5)
Does the GLM that you have fitted in part 5(a) model the data adequately?

Exercise 2: In this exercise, data are presented on the number of fractures (`Y`) that occur in the upper seams of coal mines in the Appalachian region of western Virginia. Four explanatory variables were reported:

`INB`	inner burden thickness [feet], the shortest distance between seam floor and the lower seam
`EXTRP`	percent extraction of the lower previously mined seam
`sHeight`	lower seam height [feet]
`oTime`	time [year] that the mine has been in operation

The data from 44 mines was compiled in File `mine.dat`.

a) Execute the following R command:

```
glm(Y ~ INB + EXTRP + sHeight + oTime, family=poisson, data=Mine)
```

Which model has been fitted by this R command? What are the estimated coefficients?

b) Does the residual deviance indicate that the model from part (a) is satisfactory?

c) Find approximate 95% Wald confidence intervals on the model parameters and compare them with the 95% profile confidence intervals.

Please, turn page

d) Perform a thorough residual analysis of the fitted model from part (a).

e) (optional) If necessary improve the model from part (a).

Exercise 3: The premature birth data of Exercise 3 of Worksheet Week 4 was a simplified version of this problem.

Birth weight is an important determinant of the survival prospects of premature babies with markedly poorer prospects for very low birth weight. Prediction of the probability of survival of such babies based on information collected at or soon after birth is of considerable value. Hibbard (1986) collected data on the birth weight (in grams), gestational age (in weeks), one and five minute Apgars (a score from 0 to 9) and its venous pH of 247 premature babies and observed whether the baby survived to normal release from hospital or not. Coding a 1 for survival and 0 otherwise, the first ten rows of the data in `baby.dat` are given below:

Survival	Weight	Age	Apgar1	Apgar5	pH
1	1350	32	4	7	7.25
0	725	27	5	6	7.36
0	1090	27	5	7	7.42
0	1300	24	9	9	7.37
0	1200	31	5	5	7.35
0	590	22	9	9	7.37
1	1500	32	9	9	7.29
1	1360	29	9	9	7.44
0	600	24	4	4	7.27
1	1410	30	4	5	7.35
⋮	⋮	⋮	⋮	⋮	⋮

It seems reasonable to treat the response `Survival` for a baby as independent of that for the other babies. (A change in the treatment regime in response to the data for earlier babies or mothers making multiple appearances in the study could make this assumption more tenuous but do not affect the present study.) The aim is to develop a model for the survival probability of premature babies based on these data. Since the response (`Survival`) is clearly Bernoulli distributed, we need to fit an appropriate binary regression model.

a) Fit a GLM with all explanatory variables and run a residual analysis.

b) Find transformations of the explanatory variables that might improve the fit.

c) Define the explanatory variables

```
baby$tWeight <- ifelse(baby$Weight < 1000, 0, baby$Weight-1000)
baby$tpH <- ifelse(baby$pH < 7.27, 0, baby$pH-7.27)
```

and fit the corresponding GLM using the linear predictor

```
Weight + tWeight + Age + Apgar1 + Apgar5 + pH + tpH
```

d) Based on the information of the summary output, can you simplify the linear predictor (look at coefficients of `Weight` and `tWeight` as well as of `pH` and `tpH`)

e) Run a variable selection using AIC and BIC. Run a residual analysis of the simpler model.

Exercise 4: An electric utility interested in developing a model relating peak hour demand (`Y`) to total energy usage (`x`, in kilowatt-hours) during the month. This is an important planning problem because while most customers pay directly for energy usage, the generation system must be large enough to meet the maximum demand imposed. Data for 53 residential customers for the month of August 1979 are collected in the file `eUsage.dat`.

a) Assume that the response `Y` is independently Gaussian distributed and fit a simple linear regression model. What are the estimated coefficients?

   Perform a thorough residual analysis. What are your conclusions?

b) Assume that the response `Y` is independently gamma distributed and use the identity link for fitting the GLM. What are the estimated coefficients?

   Is this model more adequate?

c) Clarify with a "GAM-Fit" whether the explanatory variable should be transformed.

d) Find approximate 95% Wald confidence intervals on the slope parameter and compare it with the corresponding 95% profile confidence interval.

e) Could the response (`Y`) be exponentially distributed? (Please note all steps of your consideration.)